

intermediate-elective

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualise missing data
library(naniar)

# arrange plots
library(patchwork)

# extra packages required for visualisation
library(ggtext)
library(showtext)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",

```

```

    site == "NMC" ~ "Main Channel",
    site == "NPB" ~ "South of Phelps Bridge",
    site == "NPB1" ~ "North of Phelps Bridge",
    site == "NPB2" ~ "Phelps Road",
    site == "NWP" ~ "West Pond",
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))

# new object created from aq_ins_clean
logsalinity2 <- aq_ins_clean |>
  # log + 1 transform mean abundance/liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(logsalinity2)

# A tibble: 1 x 24
  date_site      date                site  taxon  ave_lit med_ph med_sal med_do
  <chr>          <dtm>                <chr> <chr>   <dbl>  <dbl>  <dbl>  <dbl>
1 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2  hemip~  0.267   7.59   1.99   0.65
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

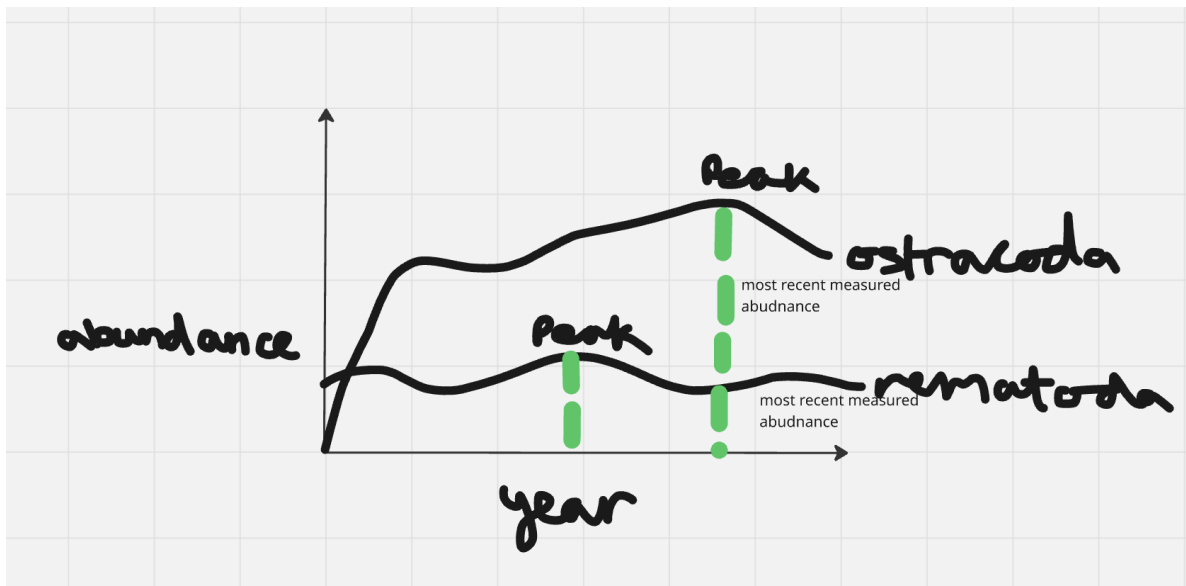
1. Explain your inspiration

Which visualization you chose for your inspiration? - I chose Steven Ponce's "Coal falls. Renewables rise."

Why it makes sense for your dataset of choice? - It makes sense as it shows how the abundance of 2 different aquatic invertebrates change through time.

which variables from your dataset will be shown in your visualization, and what visual components they map onto (axes, shapes, colors, etc.) - x-axis will be the date (year) - y-axis will be the abundance (log transformed average abundance per liter) - I will use points and lines (line graph) to show how the 2 taxons increase or decrease in abundance over time. I am choosing the taxon ostracoda and nematoda - I will also show when their abundance peaks _ I will use the same colours as the inspiration

2. Plan your figure



3. Code your figure

```
taxon_abundance <- logsalinity2 |>
  mutate(
    date = ymd(date),
    year = year(date)
  ) |>
  filter(
    taxon %in% c("ostracod", "nematode"),
    year >= 2020,
    year <= 2024
  ) |>
  group_by(year, taxon) |>
  summarise(
    abundance = mean(log_ave_lit, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(
    taxon = factor(taxon, levels = c("ostracod", "nematode"))
  )
```

```

taxon_peaks <- taxon_abundance |>
  group_by(taxon) |>
  slice_max(abundance, n = 1, with_ties = FALSE) |>
  ungroup()

```

```

taxon_end <- taxon_abundance |>
  group_by(taxon) |>
  slice_max(year, n = 1, with_ties = FALSE) |>
  ungroup()

```

```

# selecting colours for ostracods and nematodes
ostracod_gold <- "#F4A932"
nematoda_blue <- "#6A80A0"

# plot figure using taxon_abundance dataset
ggplot(taxon_abundance,
  aes(x = year,
    y = abundance,
    color = taxon,
    group = taxon)) +
  # line to connect all points
  geom_line(linewidth = 1) +
  # adding point to figure
  geom_point(size = 2.5) +
  # increasing size of peak points
  geom_point(data = taxon_peaks,
    aes(x = year, y = abundance),
    size = 4) +
  # adding dotted line for peaks
  geom_vline(data = taxon_peaks,
    aes(xintercept = year,
      color = taxon),
    linetype = "dotted",
    linewidth = 1) +
  # labelling the peaks
  geom_text(data = taxon_peaks,
    aes(label = paste0(taxon, "peak: ", year)),
    vjust = -1,
    size = 4,
    show.legend = FALSE) +
  # labelling the end points
  geom_text(data = taxon_end,

```

```

aes(label = paste0(taxon, ": ", round(abundance, 2))),
hjust = -0.1,
size = 4,
show.legend = FALSE) +
# adding a different colour for each taxon
scale_color_manual(values = c(
  "ostracod" = ostracod_gold,
  "nematode" = nematoda_blue
)) +
# y-axis and x-axis labels and titling
labs(
  title = "Ostracod and Nematode Abundance Over Time at NCOS",
  x = "Year",
  y = "Log-transformed average abundance per liter",
  color = "Taxon"
) +
# expanding x-axis so all labels fit
scale_x_continuous(expand = expansion(mult = c(0.02, 0.15)))

```

